

Relational Database Design for Academic Management System

```
create database suriya;
use suriya;
CREATE DATABASE ABCorganizations;
show databases;
USE ABCorganizations;
-- Create Faculty Table
CREATE TABLE Faculty (
FacultyID INT AUTO_INCREMENT PRIMARY KEY,
FacultyName VARCHAR(100) NOT NULL
);

-- Create Student Table
CREATE TABLE Student (
StudentID INT AUTO_INCREMENT PRIMARY KEY,
StudentName VARCHAR(100) NOT NULL
);
-- Create Department Table
CREATE TABLE Department (
DepartmentName VARCHAR(100) PRIMARY KEY,
FacultyID INT,
StudentID INT,
FOREIGN KEY (FacultyID) REFERENCES Faculty(FacultyID) ON DELETE SET
NULL,
FOREIGN KEY (StudentID) REFERENCES Student(StudentID) ON DELETE SET
NULL

);
-- Insert data into Faculty table
INSERT INTO Faculty (FacultyName) VALUES
('Dr. Smith'),
('Dr. Johnson'),
('Dr. Parker');

-- Insert data into Student table
INSERT INTO Student (StudentName) VALUES
('Alice'),
('Bob'),
('Charlie');

-- Insert data into Department table
INSERT INTO Department (DepartmentName, FacultyID, StudentID) VALUES
('Computer Science', 1, 1),
```

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('Mathematics', 2, 2),
('Physics', NULL, NULL); -- Department without associated faculty or student
-- Delete a department (e.g., 'Mathematics')
DELETE FROM Department WHERE DepartmentName = 'Mathematics';

-- Check remaining data in Faculty and Student tables (data is preserved)
SELECT * FROM Faculty;
SELECT * FROM Student;
-- Add a new faculty member who is not yet assigned to any department
INSERT INTO Faculty (FacultyName) VALUES ('Dr. Adams');

-- Add a new student who is not yet assigned to any department
INSERT INTO Student (StudentName) VALUES ('Eve');

-- Check Faculty and Student tables
SELECT * FROM Faculty;
SELECT * FROM Student;

-- Later, assign the new faculty and student to a department
UPDATE Department
SET FacultyID = 4, StudentID = 4
WHERE DepartmentName = 'Physics';

-- Verify the assignment
SELECT * FROM Department;
-- View all departments with associated faculty and students
SELECT
d.DepartmentName,
f.FacultyName AS Faculty,
s.StudentName AS Student
FROM Department d
LEFT JOIN Faculty f ON d.FacultyID = f.FacultyID
LEFT JOIN Student s ON d.StudentID = s.StudentID;

-- Check orphaned Faculty or Students (if any)
SELECT * FROM Faculty WHERE FacultyID NOT IN (SELECT FacultyID FROM
Department);
SELECT * FROM Student WHERE StudentID NOT IN (SELECT StudentID FROM
Department);

```

OUTPUT:

Database
abcorganizations
company
information_schema
mysql
orders
performance_schema
prema
rubydb
sakila
school_db
sys
world

FacultyID	FacultyName
1	Dr. Smith
2	Dr. Johnson
3	Dr. Parker
NULL	NULL

StudentID	StudentName
1	Alice
2	Bob
3	Charlie
NULL	NULL

FacultyID	FacultyName
1	Dr. Smith
2	Dr. Johnson
3	Dr. Parker
4	Dr. Adams
NULL	NULL

Result Grid			Filter Rows:
	StudentID	StudentName	
▶	1	Alice	
	2	Bob	
	3	Charlie	
	4	Eve	
*	NULL	NULL	

Result Grid				Filter Rows:	Edit:
	DepartmentName	FacultyID	StudentID		
▶	Computer Science	1	1		
	Physics	4	4		
*	NULL	NULL	NULL		

Result Grid				Filter Rows:	Export
	DepartmentName	Faculty	Student		
▶	Computer Science	Dr. Smith	Alice		
	Physics	Dr. Adams	Eve		

Result Grid			Filter Rows:
	FacultyID	FacultyName	
▶	2	Dr. Johnson	
	3	Dr. Parker	
*	NULL	NULL	

Result Grid			Filter Rows:
	StudentID	StudentName	
▶	2	Bob	
	3	Charlie	
*	NULL	NULL	